

PROJECT PROPOSAL · REVISED SCOPE

# Robust Numerical Parameter Estimation for Photovoltaic Cells

*Hybrid Root-Finding · Nonlinear Least Squares · Monte Carlo Robustness Analysis*

Extension of “Comparative Study of Parameter Extraction from a Solar Cell or a Photovoltaic Module...” — Qin et al., Energies, 2024

## TEAM

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# Base Paper & the Research Gap

## Qin et al., Energies 2024

- Extracts SDM (5-param) and DDM (7-param) parameters from measured I-V data of a solar cell and a PV module
- Combines 5 metaheuristic optimizers (AEO, GBO, GNDO, BO, RTH) with 3 current-calculation methods: Approximation, Lambert W, Newton-Raphson
- Shows NR and Lambert W give near-identical, accurate current & RMSE; the Approximation method is inaccurate
- Best reported RMSE on the RTC cell:  $\approx 7.73 \times 10^{-4}$  (SDM),  $6.94 \times 10^{-4}$  (DDM)

## WHAT REMAINS TO INVESTIGATE

*The base paper mainly evaluates metaheuristic search over parameters. We instead investigate the numerical behaviour of the complete deterministic estimation process itself — the part left unexamined.*

1

Can deterministic numerical fitting (Gauss-Newton / LM) reduce dependence on metaheuristics?

2

Can a safeguarded hybrid solver improve Newton-Raphson's robustness without sacrificing speed?

3

How stable are the extracted parameters when measured I-V data contain noise?

*This is the direct response to the earlier feedback that a solver benchmark alone lacks depth.*

# Why the PV Equation Needs a Root Solver

The single-diode model (SDM) relates output current  $I$  to voltage  $V$  through an implicit transcendental equation —  $I$  appears on both sides, once linearly and once inside an exponential:

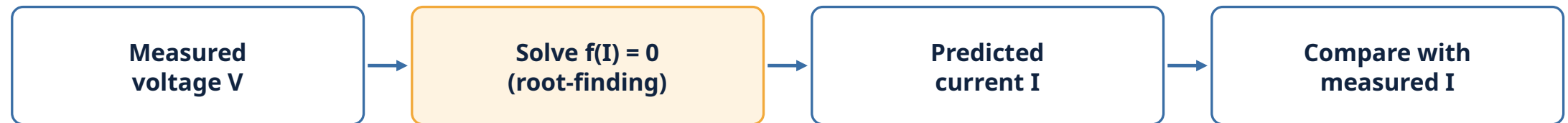
$$I = I_{ph} - I_0 \left[ \exp\left( \frac{V + I \cdot R_s}{n \cdot V_t} \right) - 1 \right] - \frac{V + I \cdot R_s}{R_{sh}}$$

## Unknown parameters, $\theta$

- $I_{ph}$  — photogenerated current
- $I_0$  — diode reverse saturation current
- $R_s, R_{sh}$  — series & shunt resistance
- $n$  — diode ideality factor

## Why it's implicit

- Given  $V$  and a trial  $\theta$ ,  $I$  cannot be isolated algebraically
- Must be solved as a root-finding problem:  $f(I) = \text{predicted} - I = 0$
- Base paper compares 3 ways to get this: Approximation, Lambert W, Newton-Raphson



# Parameter Estimation as Nonlinear Least Squares

Rather than letting a metaheuristic search blindly for  $\theta$ , we treat parameter extraction as a classic curve-fitting problem and solve it with deterministic calculus:

1

**Parameter vector**

$$\theta = [ I_{ph}, I_0, R_s, R_{sh}, n ]$$

2

**Model prediction**

For each measured  $(V_i, I_{imeas})$ , the model predicts  $I_{ipred}(\theta)$  by solving  $f(I) = 0$

3

**Residual**

$$r_i(\theta) = I_{ipred}(\theta) - I_{imeas}$$

4

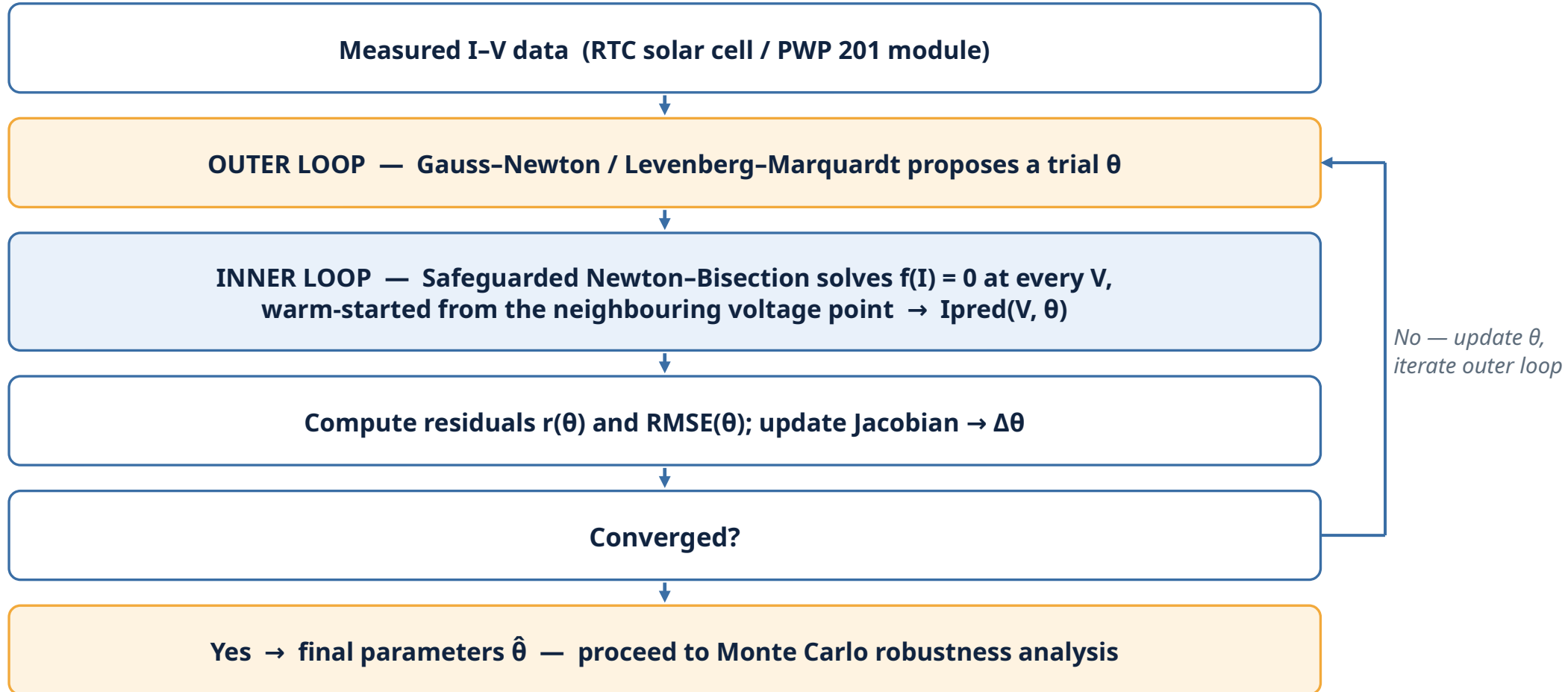
**Objective function**

$$F(\theta) = \frac{1}{2} \sum_i r_i(\theta)^2 \quad \leftarrow \text{minimize this over } \theta$$

*Because  $f(I)=0$  is itself nonlinear,  $F(\theta)$  is a nonlinear least-squares problem — solved iteratively with Gauss–Newton / Levenberg–Marquardt (Slide 6), each iteration calling the current solver (Slide 7).*

# The Complete Numerical Estimation Pipeline

A two-level iteration: an outer loop fits PV parameters; an inner loop solves for current at every trial.



# Deterministic Parameter Fitting

## Gauss-Newton

- Linearizes residuals  $r(\theta)$  around the current estimate using the Jacobian  $J$
- At each iteration, solve the normal equations for the update  $\Delta\theta$
- Then update:  $\theta_{k+1} = \theta_k + \Delta\theta$
- Fast near the solution, but can be unstable if  $J^T J$  is ill-conditioned

$$(J^T J) \Delta\theta = -J^T r$$

## Levenberg-Marquardt

- Adds a damping term  $\lambda I$  to stabilize Gauss-Newton when  $J^T J$  is poorly conditioned
- Smoothly blends gradient-descent (large  $\lambda$ , safe) and Gauss-Newton (small  $\lambda$ , fast) behaviour
- Standard choice for diode-parameter fitting in the literature — not a made-up extension

$$(J^T J + \lambda I) \Delta\theta = -J^T r$$

### Course connection: solving the linear system

Both normal-equation systems above are solved with Gaussian elimination / LU decomposition — directly reusing a core numerical-methods topic, applied here to a real nonlinear-fitting problem.

*We implement and compare both, tracking convergence speed, stability, and final RMSE achieved on  $\theta$ .*

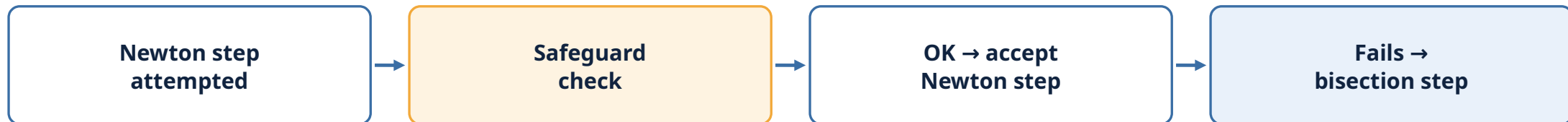
# Adaptive Hybrid Current Solver

## Safeguarded Newton-Bisection

- Start from a physically meaningful bracket:  $0 \leq I \leq I_{sc}$
- Attempt a standard Newton step:  $I_{k+1} = I_k - f(I_k)/f'(I_k)$
- Fall back to a bisection step whenever: the derivative is near zero, Newton jumps outside the bracket, or the residual worsens
- Result: Newton's speed + bisection's guaranteed reliability

## Warm-Starting / Continuation

- Measured voltages are ordered:  $V_1, V_2, \dots, V_N$
- Neighbouring points on an I-V curve have similar currents, so  $I(V_i)$  is an excellent initial guess for  $I(V_{i+1})$
- Continuation rule:  $I_0(V_{i+1}) = I(V_i)$  — instead of restarting from a fixed guess (e.g.  $I_0 = 0.5$ ) every time
- Compare: standard NR vs. warm-start NR vs. warm-start safeguarded NR, on total iterations & runtime



*This inner solver runs at every voltage point, for every trial  $\theta$  the outer loop proposes — its speed and reliability directly determine total pipeline runtime.*

# Monte Carlo Analysis Under Measurement Noise

Real I-V measurements are never exact. We inject synthetic Gaussian noise into the measured current and re-run the full estimation pipeline many times:

$$\tilde{I_i} = I_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$$

Test protocol

- Noise levels: 0.1%, 0.5%, 1%, 2% of measured current
- 100–500 independent trials per noise level
- Full parameter extraction re-run every trial

Question answered

- How reliable are the extracted PV parameters when the experimental I-V measurements contain realistic noise?
- A genuine, actively studied issue in PV parameter-estimation literature

Reported per noise level:

| Noise | Iph variation | Rs variation | Rsh variation | RMSE | Failure rate |
|-------|---------------|--------------|---------------|------|--------------|
| 0%    | ...           | ...          | ...           | ...  | ...          |
| 0.5%  | ...           | ...          | ...           | ...  | ...          |
| 1%    | ...           | ...          | ...           | ...  | ...          |
| 2%    | ...           | ...          | ...           | ...  | ...          |



# Stretch Goals (If Time Permits)

Core deliverables are Slides 5–8. The following extend the numerical depth further and will be pursued opportunistically.

## A SDM vs. DDM — A Numerical Comparison

- Single-diode model: 5 parameters. Double-diode: 7 — lower RMSE, but costlier to fit.
- Base paper: RTC RMSE  $\approx 7.73 \times 10^{-4}$  (SDM) vs  $6.94 \times 10^{-4}$  (DDM)
- Question: is the added numerical complexity of DDM justified by its accuracy gain?
- Compare: RMSE, runtime, iteration count, convergence failures, sensitivity to initial values, robustness under noise

## B Condition Number & Identifiability

$\kappa(J^T J)$  — computed for free during Gauss–Newton

- A large  $\kappa(J^T J)$  means small measurement errors can cause large swings in the fitted parameters.
- Explains an otherwise puzzling result: RMSE stays flat while  $R_{sh}$  changes significantly.
- Key distinction: good curve fitting  $\neq$  accurately identified physical parameters.
- PV parameter identifiability is an established open problem in the literature — this ties our project to it directly.

# How We Will Measure Success

## 1 Accuracy

- RMSE
- Parameter error
- I-V curve fit quality

## 2 Efficiency

- Runtime
- Iteration count
- Function evaluations

## 3 Stability

- Failure rate
- Initial-guess sensitivity
- Noise sensitivity

### EXPECTED CONTRIBUTION

**A faster and more robust deterministic numerical parameter-estimation pipeline for PV models, together with a clear analysis of when and why the numerical methods succeed or fail.**

*Committed scope: Gauss–Newton / LM parameter fitting + Safeguarded Newton–Bisection current solver + Monte Carlo robustness analysis.*

*Optional extensions: SDM vs. DDM comparison, condition-number / identifiability analysis.*